Neuroinflammation and blood-brain barrier permeability in irritable bowel syndrome: a [^{18}F]DPA-714 PET study

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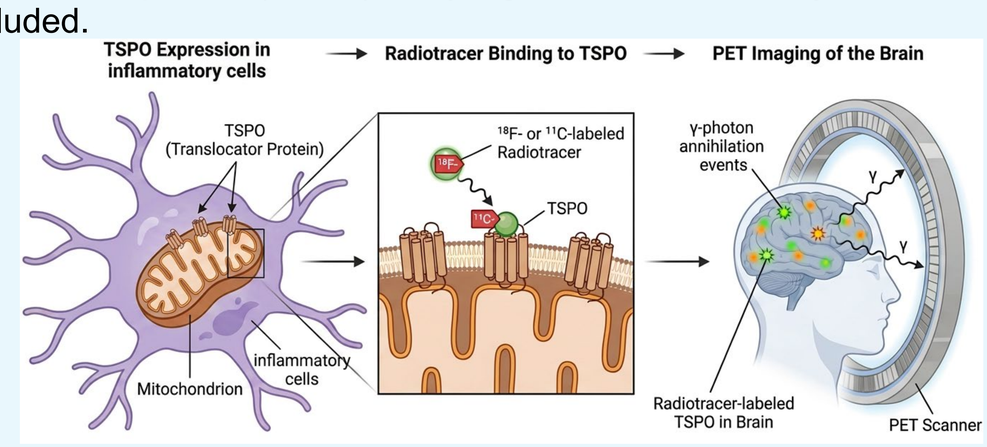
1. INTRODUCTION & AIMS

- **Irritable bowel syndrome (IBS)** is often accompanied by extraintestinal symptoms such as fatigue, pain, anxiety, and depression.
- These extraintestinal symptoms may be driven by chronic low-grade immune activation, which may propagate to the brain (**neuroinflammation**), particularly if blood-brain barrier (BBB) integrity is compromised.
- **Aims:**

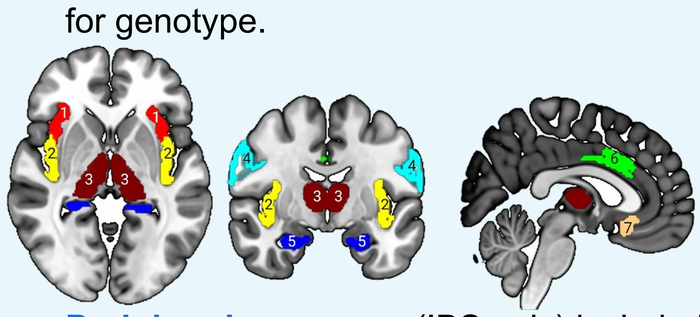
1) Investigate differences in neuroinflammation and BBB permeability between individuals with IBS and healthy controls (HC).
2) Investigate the relationship between neuroinflammation and BBB permeability with comorbidity and IBS symptom severity in IBS.
3) Investigate the relationship between neuroinflammation and BBB permeability with immune markers and intestinal permeability in IBS.
4) Investigate neuroinflammation and BBB permeability as mediators of the peripheral measures – symptom/comorbidity severity relationship.

2. MEASURES

- **Sample: 16 individuals with IBS** (50% women, mean age = 43.7 years) and **16 healthy controls** (HC; 56% women, mean age = 44.2 years) participated in a PET scan using the [^{18}F]DPA-714 radiotracer binding to translocator protein (TSPO). Only high- and mixed-affinity binders were included.



- **Total distribution volume (V_T),** the quantification of TSPO availability derived from full kinetic modeling, was used as a proxy for **neuroinflammation.**
- **Plasma-to-brain tracer influx (K_1)**, a parameter contributing to V_T estimation, was used as a proxy for **BBB permeability.**
- Both were measured in 7 bilateral regions of interest (ROIs) and residualized for genotype.

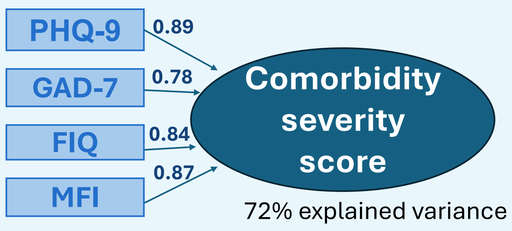


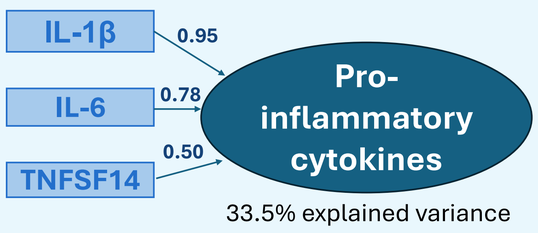
1. Anterior insula
2. Posterior insula
3. Thalamus
4. Primary somatosensory cortex
5. Hippocampus
6. Anterior midcingulate cortex
7. Subgenual anterior cingulate cortex

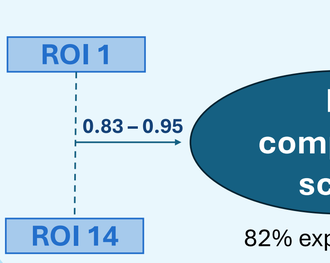
- **Peripheral measures** (IBS only) included five **circulating cytokines** (IL-1β, IL-6, TNFSF14, IL-1RN, IL-33) and two **intestinal permeability** markers (serum claudin-1, LBP).
- **Questionnaires** assessed **comorbidity severity** (PHQ-9: depression, GAD-7: anxiety, MFI: fatigue, FIQ: pain) and **IBS symptom severity** (IBS Severity Scoring System, IBS-SSS).

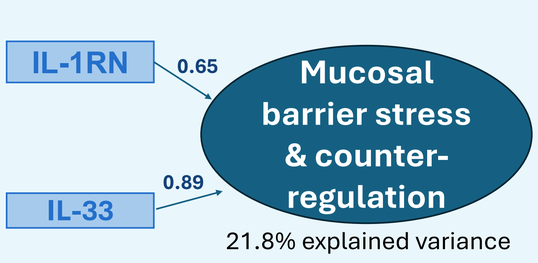
3. DATA REDUCTION

- Individual component scores, derived from **principal component analyses** with varimax rotation, were used to generate composite measures of **comorbidity severity, cytokines,** and **K_1 across ROIs.**



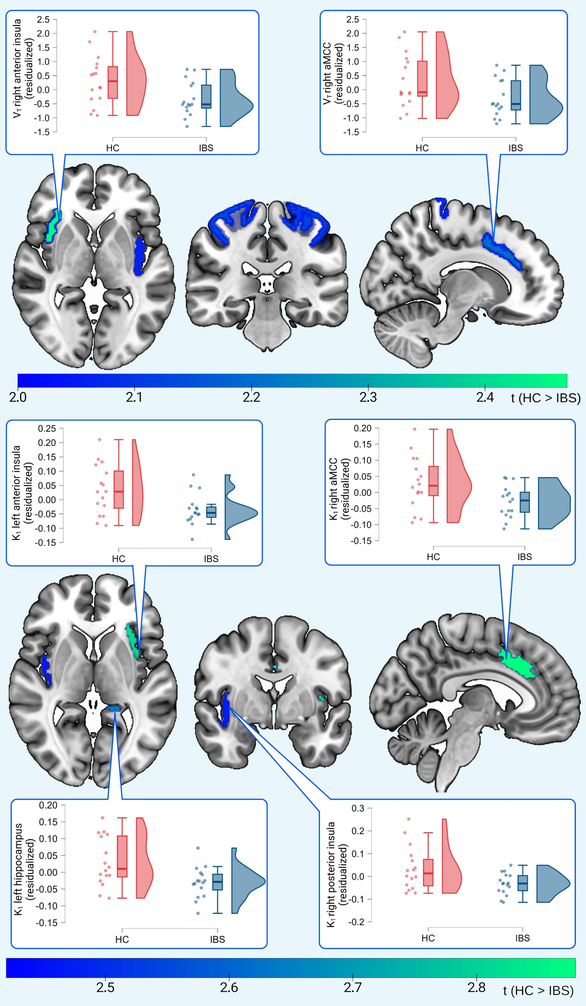






4. RESULTS

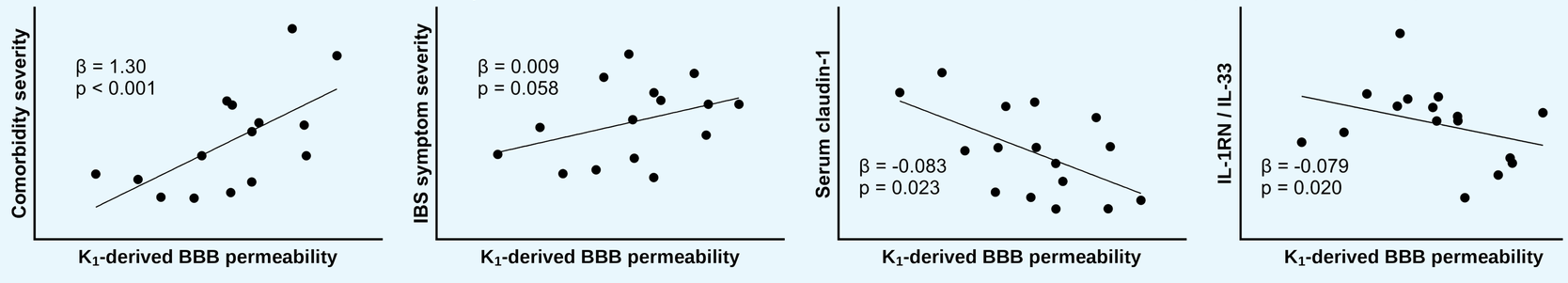
Individuals with IBS had *lower* V_T and *lower* K_1 values ($a_{FDR} < 0.05$)

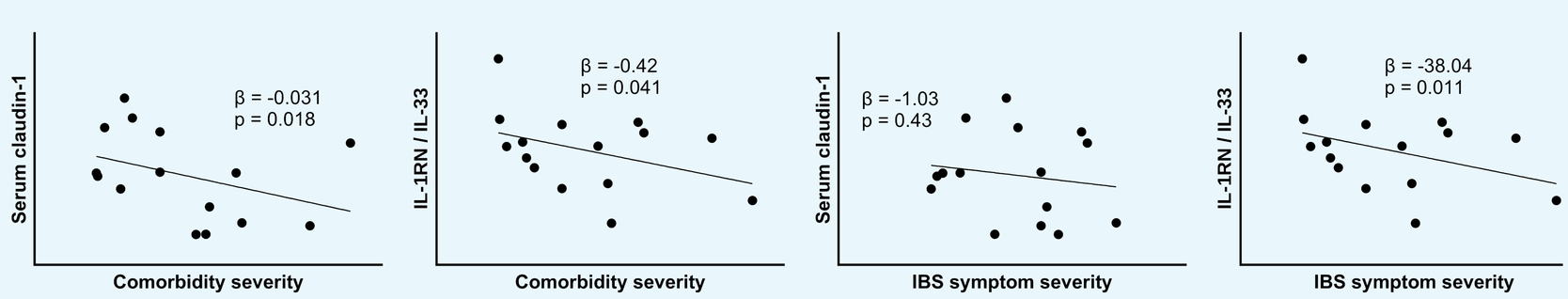


Top: V_T, bottom: K_1; values residualized for TSPO genotype.

Within the IBS group:

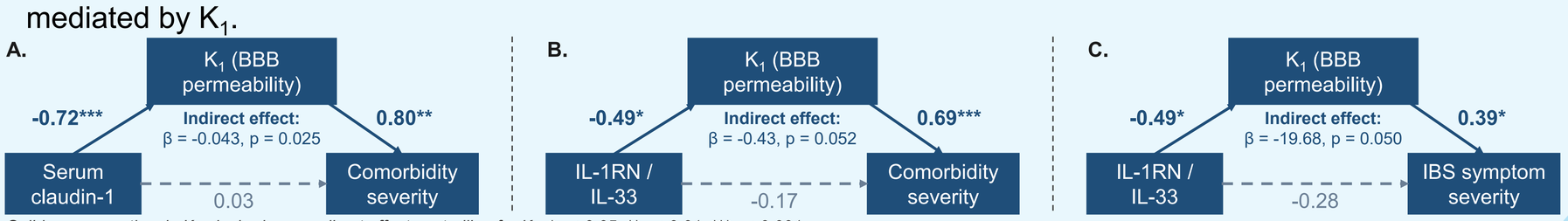
- Increased K_1 was associated with increased comorbidity and IBS symptom burden.
- Higher serum claudin-1 levels and higher scores on the mucosal barrier stress & counter-regulation cytokine factor were associated with *decreased* K_1, as well as with decreased comorbidity and IBS symptom severity.





Associations within the IBS group (n = 16); values residualized for TSPO genotype.

- The negative relationships between these peripheral markers and comorbidity / IBS symptom severity were fully mediated by K_1.



*Solid arrows: paths via K_1; dashed arrow: direct effect controlling for K_1. *p < 0.05, **p < 0.01, ***p < 0.001.*

5. CONCLUSION

- In contrast to our expectation, the study demonstrated **lower TSPO availability**, as a marker of neuroinflammation, and **lower K_1-derived BBB permeability in individuals with IBS** compared to HC.
- BBB permeability was related to both clinical symptom burden and peripheral markers.
- This points to a **complex role of BBB function** in linking peripheral physiology and centrally driven symptoms **in IBS.**
- Given the caveats of TSPO PET and the modest sample size, these findings must be interpreted with caution.

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